AIML LABORATORY — MACHINE LEARNING MINI PROJECT

AutoValue AI

Used Car Price Prediction and Deal Analysis Using Machine Learning

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Summary. Three regression models were trained on 5,534 historical CarDekho listings and compared with 5-fold cross-validation. The selected model, **Linear Regression**, achieved a test **MAE of ₹84,331**, RMSE of ₹2,13,694 and **R^2 of 0.805** on 1,384 held-out cars. A Streamlit app estimates a car's resale price and compares it with an asking price.

1. Aim

To estimate the resale price of a used car from its characteristics with a machine-learning regression model, and to help a buyer compare that estimate with the seller's asking price.

2. Problem statement

A used car's value depends on its model, age, usage, fuel type, transmission and ownership history. A buyer often can't tell whether an asking price is reasonable. This project learns the relationship between these attributes and historical selling prices, and uses it to estimate prices for new inputs. Because the target (price) is continuous, this is a **supervised regression** problem.

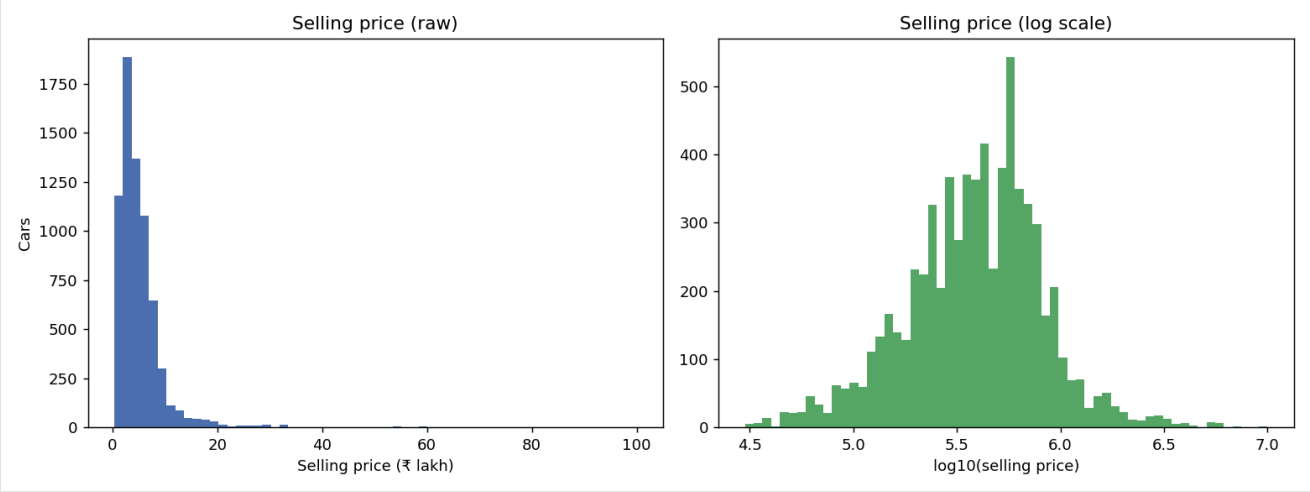
3. Dataset

CarDekho *Car details v3* (Kaggle, nehalbirla): 8,128 listings × 13 columns, manufacturing years 1983–2020, prices ₹29,999–₹1 crore.

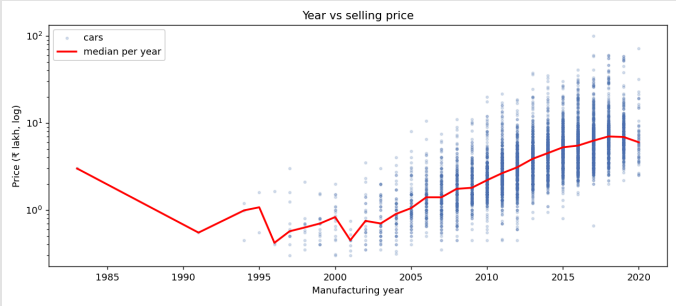
- **Cleaning:** 1,203 exact duplicates removed *before* splitting (to prevent train/test leakage); 5 "Test Drive Car" rows excluded; 2 implausible odometer readings (>10 lakh km) removed → **6,918 rows**.
- **Inputs vs features:** the 7 original input fields are `name, year, km_driven, fuel, seller_type, transmission, owner`. `name` has 2,058 variants, mostly with very few rows, so it is split into `brand` (32) and `model` (212). The model therefore uses **8 features**.

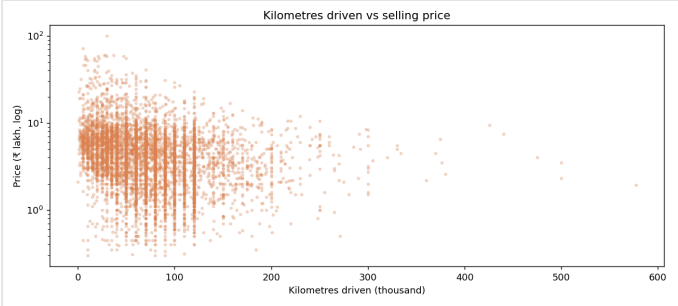
> **Data age:** the data ends in 2020. Estimates are historical, not current-market valuations, and the app only accepts manufacturing years in the training range (1983–2020).

4. Exploratory data analysis

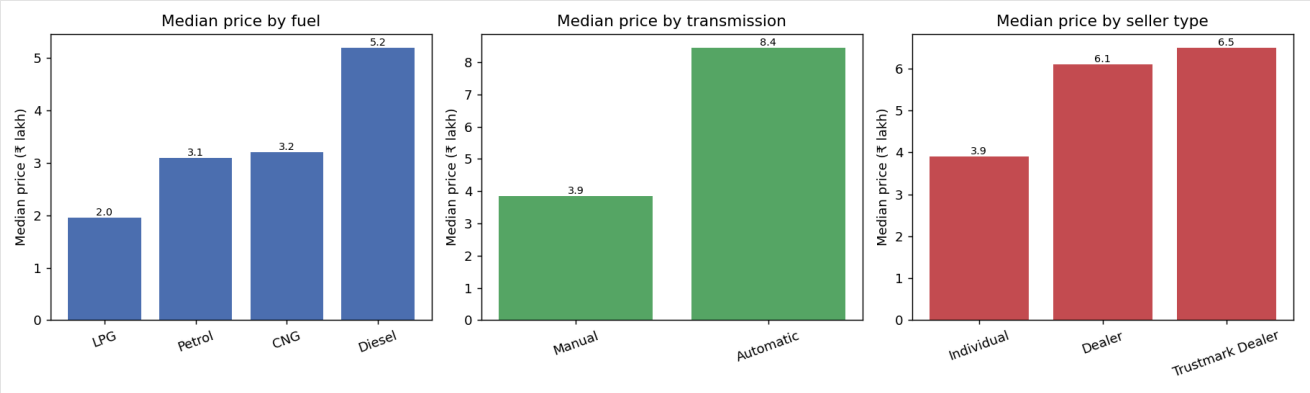


Prices are strongly right-skewed (skewness ≈ 5.4) but nearly symmetric on a log scale, so models are trained on log(price) and predictions converted back to rupees.





Newer cars sell for more (correlation of year with log price ≈ 0.72). Kilometres driven has a weaker negative association (≈ −0.23) and is itself correlated with age.



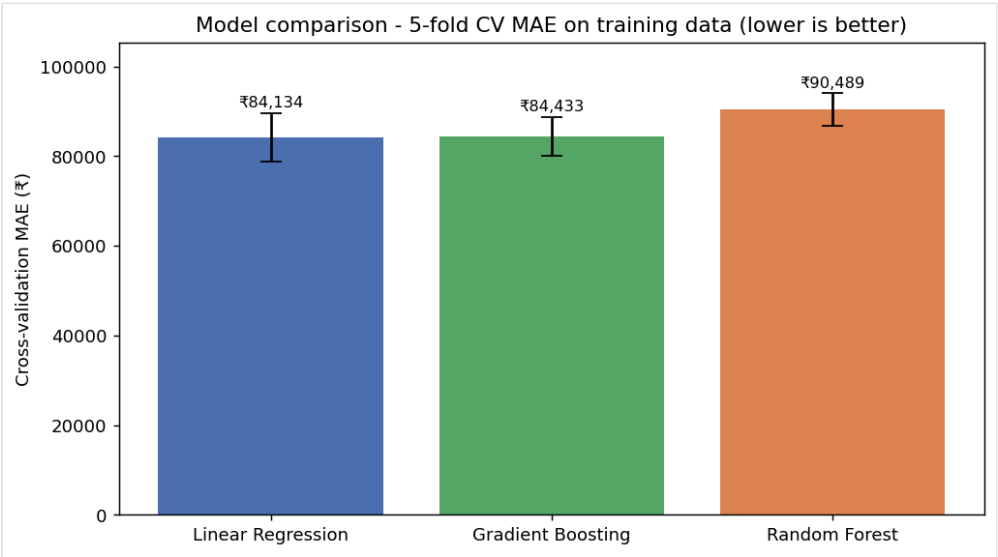
Automatic, diesel and dealer-listed cars have higher median prices. These are associations, not causal effects.

5. Methodology

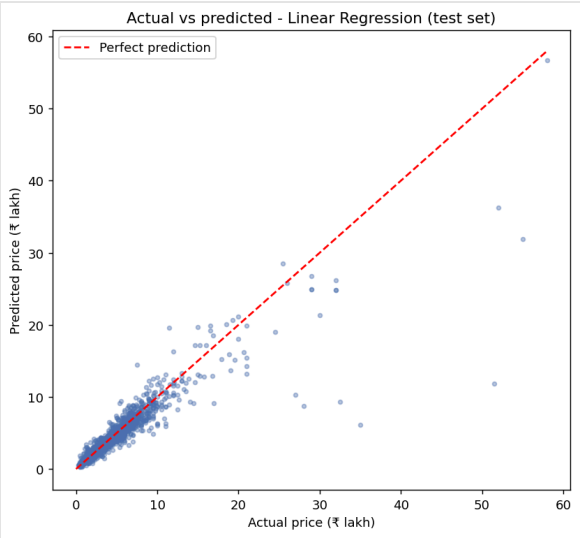
- **Preprocessing** inside a scikit-learn pipeline, fitted on training data only: one-hot encoding of brand, model, fuel, seller type, transmission and owner (unknown categories ignored, rare ones grouped); numeric year and km (scaled for Linear Regression).
- **Target:** log(price) via `TransformedTargetRegressor`; all metrics are reported in rupees.
- **Models:** Linear Regression (baseline), Random Forest and Gradient Boosting, each with a small hyper-parameter grid.
- **Selection:** 80/20 split (seed 42) → the same 5-fold cross-validation on the training set for all models → lowest CV MAE → single evaluation on the test set. Test data was never used for tuning or selection.
- **Metrics:** MAE (average absolute error), RMSE (penalises large errors), R^2 (variance explained).

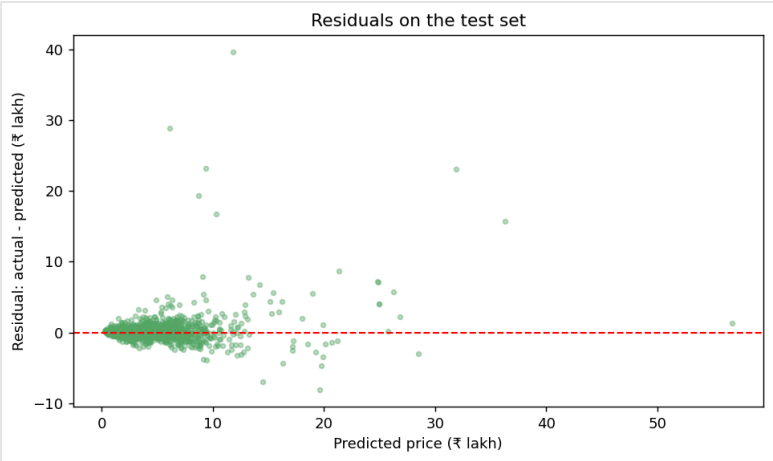
6. Results

Model	CV MAE	CV R^2	Test MAE	Test RMSE	Test R^2
Linear Regression	**₹84,134 ± 5,386**	**0.825**	**₹84,331**	**₹2,13,694**	**0.805**
Gradient Boosting	₹84,433 ± 4,285	0.800	₹84,772	₹2,28,885	0.777
Random Forest	₹90,489 ± 3,635	0.776	₹92,388	₹2,38,679	0.757



Linear Regression had the lowest cross-validation MAE and was selected. Its lead over Gradient Boosting is far smaller than the CV spread, so the two are effectively tied. Linear Regression is also the simplest model to explain. On log(price), with brand and model encoded, it behaves like a multiplicative depreciation model, which suits this data. Random Forest was the weakest.



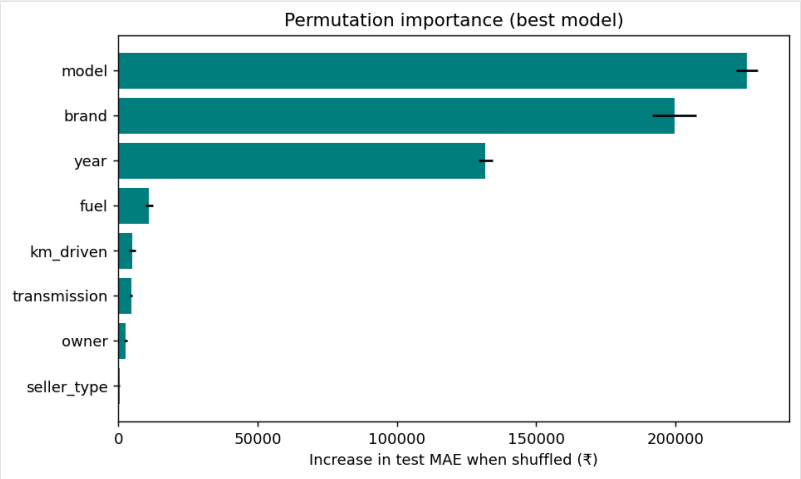


Predictions follow the actual prices closely for cheap and mid-range cars. Errors grow for expensive cars, which is why RMSE (₹2,13,694) is much larger than MAE (₹84,331).

Robustness check — unseen car models

With cross-validation grouped by car model (each validation fold contains only models absent from that fold's training data), MAE rises to **₹2,18,995** and R^2 falls to **0.464**. The headline result applies to car models represented in the training data. The app only offers known models and warns when a model has few examples.

Feature importance



Feature	MAE increase when shuffled
model	₹2,25,512
brand	₹1,99,492
year	₹1,31,827
fuel	₹11,099
km_driven	₹5,192
transmission	₹4,621
owner	₹2,739
seller_type	₹464

Permutation importance on the full pipeline shows that car **model**, **brand** and **year** drive the predictions. The other fields add relatively little once those are known.

7. Application

A Streamlit app loads the saved pipeline (it doesn't retrain) and has three tabs:

- **Price Estimator:** brand → model, year, km, fuel, seller type, transmission, ownership; inputs are validated (e.g. negative km or a year outside 1983–2020 is rejected with a message). The estimate is shown prominently, followed by a separate note on the test MAE. Any MAE-based range is labelled illustrative, not a statistical prediction interval.
- **Deal Checker:** % difference = (asking − estimate) / estimate × 100; labelled *Potentially below estimate* (< −5%), *Close to estimate* (±5%) or *Potentially above estimate* (> +5%). The thresholds are a project design choice.
- **Model Insights:** metrics, comparison, robustness check, charts and limitations.

The app was checked with 14 unit tests (validation, deal arithmetic, rupee formatting, prediction) and a scripted smoke test covering a real prediction, equal/higher/lower asking prices, negative km and an unsupported year.

8. Limitations

- Historical listings up to 2020 only; no current market data.
- Condition, accident and service history, and location are not modelled.
- Rare models and small categories (CNG, LPG, Trustmark Dealer) have few examples.
- Estimates are educational, not valuations or purchase advice.

9. Conclusion

AutoValue AI implements a complete regression workflow: cleaning with leakage-safe duplicate removal, feature engineering of brand and model from the car name, a fair cross-validated comparison of three algorithms, and a single held-out evaluation. Linear Regression was selected, reaching a test MAE of ₹84,331 and R^2 of 0.805. The grouped check shows that reliability depends on the car model being represented in the data. The Streamlit app turns the model into a usable price estimator and deal checker, with clear statements of its limitations.